

Adaptive and Efficient Vision Model Selection using Zero-Cost and Resource-Aware Techniques

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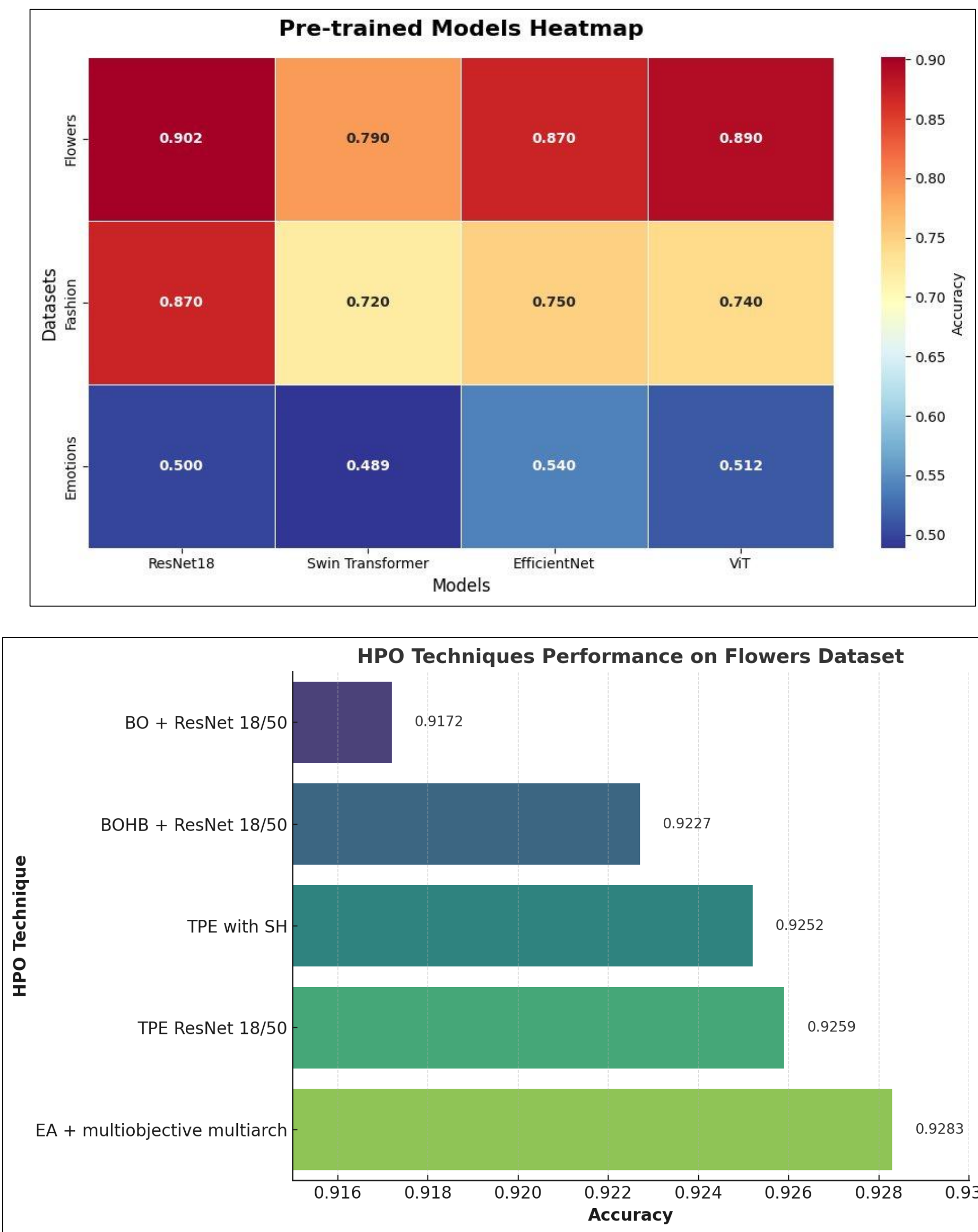
GenZs

Vision Modality

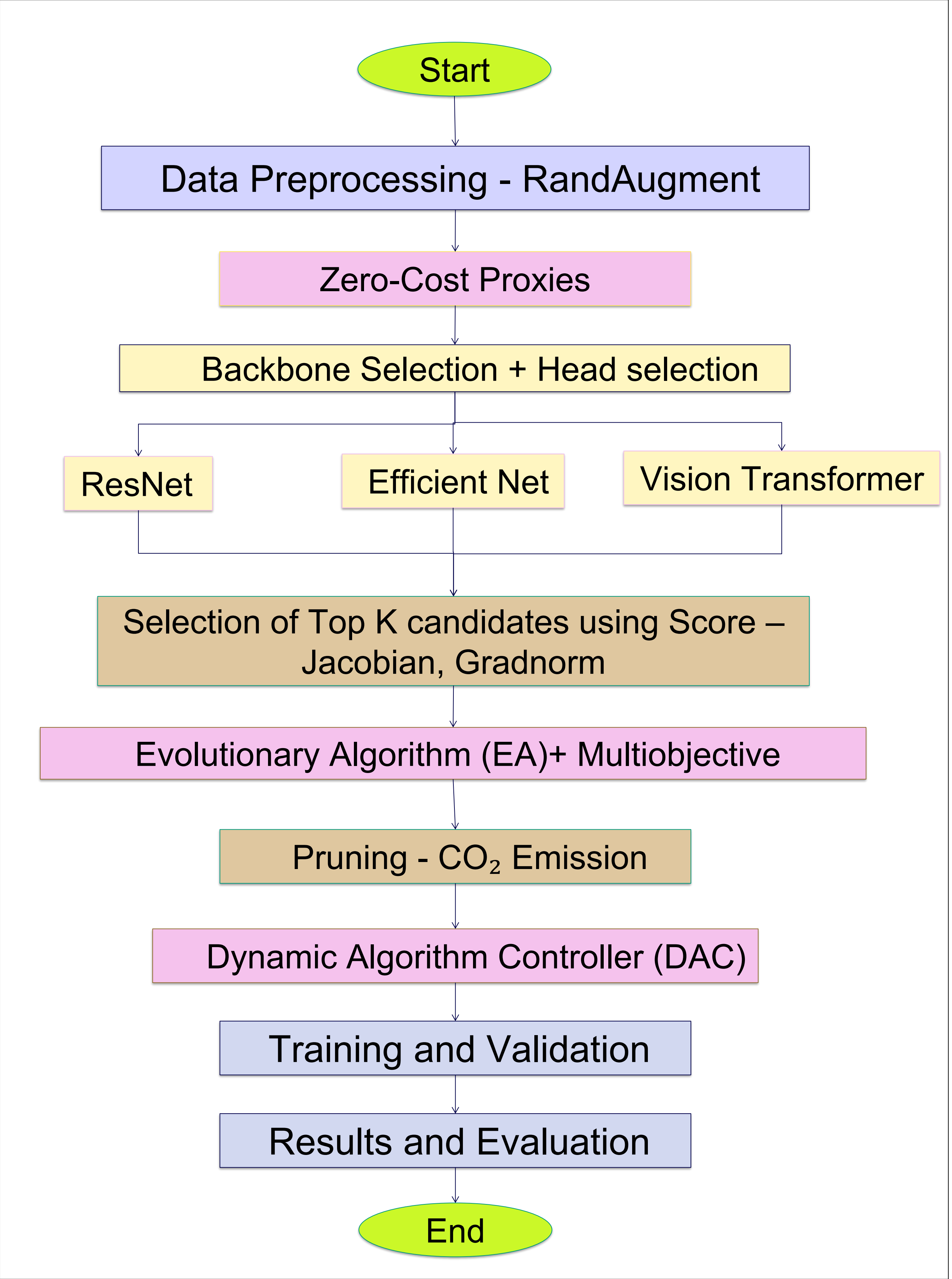
Motivation

- **Data Augmentation** (RandAugment) to balance dataset classes.
- Use **zero-cost proxies** to rapidly estimate model performance.
- Apply **evolutionary search** (Optuna NSG-III sampler) over diverse vision backbones (**ResNet**, **EfficientNet**, **Vision Transformer**)
- Select the best candidates using **pareto fronts**.
- Guide HPO through **CO2 aware strategies** for resource efficiency.
- Improve training dynamically with **DAC**- Dynamic Algorithm Configuration on **Learning Rate** and **Optimizer**.

Model Selection Strategy



Model Architecture



Reference

- [Abdelfattah, Mohamed & Mehrotra, Abhinav & Dudziak, Łukasz & Lane, Nicholas. \(2021\). Zero-Cost Proxies for Lightweight NAS.](#)
- [Benoit Courty, 'mlco2/codecarbon: v2.4.1'. Zenodo, May 10, 2024.](#)

- Week 1
- Week 2
- Week 3
- Week 4
- Week 5
- Week 6
- Week 7
- Week 8
- Week 9
- Week 10
- Bonus
- Literature

Resources Used

For Train and Development:
- GPUP100
- T4 - GPU
- CPUs

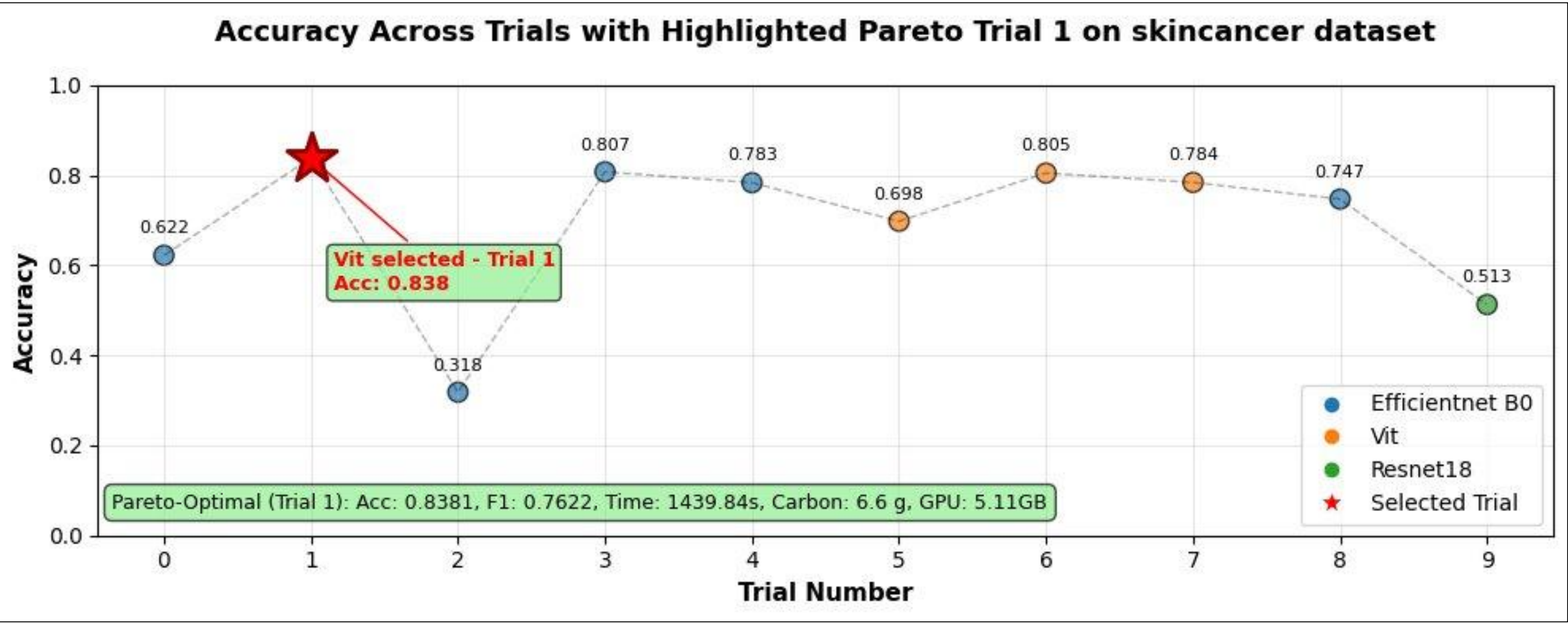
GPU Hours:
- Test Predictions : 3 hours

Workforce:
- 3 Full weeks

Number of queries for test score generation: 1

Results

Performance Metrics Across Datasets And Seeds					
Dataset	Metric	Baseline Accuracy	Seed 6	Seed 42	Seed 92
flowers	Test Accuracy	0.55	0.94	0.93	0.97
	F1		0.93	0.98	0.96
emotions	Test Accuracy	0.41	0.68	0.65	0.68
	F1		0.67	0.64	0.66
fashion	Test Accuracy	0.88	0.95	0.93	0.94
	F1		0.95	0.93	0.94
skincancer	Test Accuracy	0.71	N/A	0.86	N/A
	F1			0.79	



Limitations

1. Estimated Carbon Costs for each architecture are heuristic based, runtime variability might prune trials prematurely.
2. The use of custom heads may introduce misalignment with proxy assumptions.

Contributions

Manasi	Resnet18, Efficient net B0, Vision Transformer, DAC, Poster Design, Result Plots.
Reeya	Zero Proxies, Multi-Objective, Optuna with NSG-III Sampler, Code commenting, readme.md.
Chandrika	CO2 Strategies and Rand Augment, Code integration and Structuring, Result Plots.

